Crossbow: Webhook Delivery Service

Link to the live, deployed application:

Step-by-step instructions on how to set up and run the application locally using Docker:

Explanation of your architecture choices

Discussion on database schema and indexing strategies

Sample curl commands:

Register new user:

```
curl -X POST http://127.0.0.1:8000/api/register/ \
-H "Content-Type: application/json" \
-d '{"username": "testuser4", "email": "test@example.com", "password": "testpass123"}'
```

```
Vedansh@DESKTOP-B0CT642 MINGW64 /d/Vedansh/IITBh/Projects/crossbow
$ curl -X POST http://127.0.0.1:8000/api/register/ \
-H "Content-Type: application/json" \
-d '{"username": "testuser4", "email": "test@example.com", "password": "testp
ass123"}'
{"token":"ab5b3cce6a48b3d46c1f4899c923e1f2bfe6a40c","user_id":4,"email":"test
@example.com"}
```

The subscription id is 4 (highlighetd in red)

Login (get token)

```
curl -X POST http://127.0.0.1:8000/api/auth/token/ \
-H "Content-Type: application/json" \
-d '{"username": "testuser4", "password": "testpass123"}'
```

```
Vedansh@DESKTOP-B0CT642 MINGW64 /d/Vedansh/IITBh/Projects/crossbow
$ curl -X POST http://127.0.0.1:8000/api/auth/token/ \
-H "Content-Type: application/json" \
-d '{"username": "testuser4", "password": "testpass123"}'
{"token":"ab5b3cce6a48b3d46c1f4899c923e1f2bfe6a40c","user_id":4,"email":"test
@example.com"}
```

Use the token in the subsequent CURL commands, henceforth highlighted in green

First, create a subscription

The target url should be : http://httpbin.org/post
hence forth highlighetd in yellow

curl -X POST http://127.0.0.1:8000/api/subscriptions/ \
-H "Authorization: Token ab5b3cce6a48b3d46c1f4899c923e1f2bfe6a40c" \
-H "Content-Type: application/json" \
-d '{"subscription_id": "4", "target_url": "http://httpbin.org/post", "secret_key": "test-secret"}'

```
Vedansh@DESKTOP-B0CT642 MINGW64 /d/Vedansh/IITBh/Projects/crossbow
$ curl -X POST http://127.0.0.1:8000/api/subscriptions/ \
-H "Authorization: Token ab5b3cce6a48b3d46c1f4899c923e1f2bfe6a40c" \
-H "Content-Type: application/json" \
-d '{"subscription_id": "4", "target_url": "http://httpbin.org/post", "secret
_key": "test-secret"}'
{"id":2,"user":{"id":4,"username":"testuser4","email":"test@example.com"},"su
bscription_id":"4","target_url":"http://httpbin.org/post","secret_key":"test-
secret","created_at":"2025-05-11T10:57:50.792134Z"}
```

Send a webhook (this will now return a delivery_id)

curl -X POST http://127.0.0.1:8000/api/ingest/4/ \
-H "Authorization: Token ab5b3cce6a48b3d46c1f4899c923e1f2bfe6a40c" \
-H "Content-Type: application/json" \
-d '{"event": "test_event", "data": {"key": "value"}}'

```
Vedansh@DESKTOP-B0CT642 MINGW64 /d/Vedansh/IITBh/Projects/crossbow
$ curl -X POST http://127.0.0.1:8000/api/ingest/4/ \
-H "Authorization: Token ab5b3cce6a48b3d46c1f4899c923e1f2bfe6a40c" \
-H "Content-Type: application/json" \
-d '{"event": "test_event", "data": {"key": "value"}}'
{"message":"Webhook queued for processing","delivery_id":12}
```

This returned the delivery id as 12 (henceforth highlighted in orange)

Use this delivery_id to check the status:

curl -X GET http://127.0.0.1:8000/api/delivery-status/12/ \
-H "Authorization: Token ab5b3cce6a48b3d46c1f4899c923e1f2bfe6a40c"

```
Vedansh@DESKTOP-B0CT642 MINGW64 /d/Vedansh/IITBh/Projects/crossbow
$ curl -X GET http://127.0.0.1:8000/api/delivery-status/12/ \
-H "Authorization: Token ab5b3cce6a48b3d46c1f4899c923e1f2bfe6a40c"
{"id":12,"subscription":{"id":2,"user":{"id":4,"username":"testuser4","email"
:"test@example.com"},"subscription_id":"4","target_url":"http://httpbin.org/p
ost","secret_key":"test-secret","created_at":"2025-05-11T10:57:50.792134Z"},"
attempt_number":1,"status":"Success","http_status_code":200,"error_details":n
ull,"timestamp":"2025-05-11T10:58:43.223830Z"}
```

Delivery attempt history for a specific delivery task :

curl -X GET http://127.0.0.1:8000/api/delivery-attempt-history/12/ \
-H "Authorization: Token ab5b3cce6a48b3d46c1f4899c923e1f2bfe6a40c"

```
Vedansh@DESKTOP-B0CT642 MINGW64 /d/Vedansh/IITBh/Projects/crossbow
$ curl -X GET http://127.0.0.1:8000/api/delivery-attempt-history/12/ \
-H "Authorization: Token ab5b3cce6a48b3d46c1f4899c923e1f2bfe6a40c"
[{"id":12,"subscription":{"id":2,"user":{"id":4,"username":"testuser4","email
":"test@example.com"},"subscription_id":"4","target_url":"http://httpbin.org/
post","secret_key":"test-secret","created_at":"2025-05-11T10:57:50.792134Z"},
"attempt_number":1,"status":"Success","http_status_code":200,"error_details":
null,"timestamp":"2025-05-11T10:58:43.223830Z"}]
```

Recent delivery attempts for a specific subscription:

curl -X GET http://127.0.0.1:8000/api/subscriptions/4/recent_deliveries/ \
-H "Authorization: Token ab5b3cce6a48b3d46c1f4899c923e1f2bfe6a40c"

Estimation of the monthly cost: